IEDA3250 Stochastic Models

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Contents

1. Introduction ..... 1.
   1.1. Where Randomness Enters Engineered Systems ..... 3.
   1.2. Decisions under Uncertainty: Why Stochastic Models Matter ..... 6.
      1.2.1. Case study: Waiting-Time Paradox ..... 9.
      1.2.2. Case Study: Phantom Traffic Jam ..... 13.
      1.2.3. Case Study: Pop Mart Blind-Box ..... 16.
   1.3. Stochastic Model: Basic Concepts ..... 19.
   1.4. Overview of Topics ..... 23.
2. Markov Chains ..... 24.
   2.1. Simple Random Walk ..... 24.
      2.1.1. Definition of Simple Random Walk ..... 28.
      2.1.2. Properties of Simple Random Walk ..... 30.
   2.2. Discrete-Time Markov Chains ..... 44.
      2.2.1. From Random Walk to Markov Chains ..... 45.
      2.2.2. Chapman–Kolmogorov Equations: Multi-Step Transitions ..... 61.

2.2.3. Limiting behavior of DTMCs ..... 74.

1. Introduction

In modern Industrial Engineering practice, the central design challenge is to "optimize the system." A decision maker must:

- allocate limited resources
- while balancing competing objectives.

> ***i*** **Connection to IEDA3010 Prescriptive Analytics**
>
> IEDA3010 translates real decision problems into optimization models—identify objectives and constraints, formulate linear/network/integer/dynamic programs, and solve for optimal policy.

A typical constraint we see in IEDA3010 is a **budget constraint** $\sum_i c_i y_i \leq B$.

- **Interpretation**: producing quantity y_i incurs unit cost c_i; the total spend across all decisions cannot exceed the budget B.

⚠ Cost uncertainty in practice

In practice, unit costs c_i are *not fixed*.

- They can vary with spot prices, surge incentives, supplier lead times, and congestion.
- A deterministic budget constraint $\sum_i c_i y_i \leq B$ is often only an approximation; the true spend is random and may violate B *unless we plan for risk*.

Stochastic Models extends the prescriptive workflow to environments where the *inputs* and *system evolution* are **random**:

- demand fluctuates, service times vary, failures occur, and travel times are uncertain.

As a result, the "right" decision is no longer optimized against a single deterministic scenario; it is optimized against *risk*, *service levels*, and *long-run performance*.

1.1. Where Randomness Enters Engineered Systems

ChatGPT (LLM platform) — Example 1.1

- **Context**: an interactive, consumer-facing chat product that **streams** tokens. Each request has (i) a **prefill** stage that processes the full prompt and builds the KV-cache, then (ii) a **decode** stage that generates tokens one-by-one.
- **Resource**: GPU compute + memory (KV-cache), concurrency / queue capacity.
- **KPI**: Time to First Token (TTFT); Time Per Output Token (TPOT); p95/p99 end-to-end latency; error / rate-limit rejects.
- **Decision**: admission control / rate limits; model routing (small vs large); batching & scheduling; token caps / truncation; autoscaling.

- **Randomness**: bursty query arrivals; heterogeneous prompt lengths; variable output length (tokens); system load.

Foodpanda (on-demand marketplace) Example 1.2

- **Context**: an on-demand marketplace matching customers, merchants (restaurants/shops), and couriers in real time; service is naturally **two-stage** (merchant prep + courier pickup/delivery) and demand peaks are strongly time-of-day and location dependent.
- **Resource**: fleet size; driver hours; dispatch capacity; staging locations.
- **KPI**: pickup-time p90/p95; on-time probability; utilization; empty miles.
- **Decision**: dynamic pricing / incentives; repositioning; batching/stacking orders; service-region throttling.

- **Randomness**: stochastic trip requests; travel-time variability (incidents/weather); spatio-temporal imbalance; cancellations/no-shows.

Zara (fast fashion inventory management) Example 1.3

- **Context**: new styles launch frequently; each style has a short selling window; stores receive frequent shipments from a centralized network.
- **Resource**: limited store shelf/fixture space; DC throughput; nearshore production capacity; weekly shipment slots.
- **KPI**: sell-through in first 7–14 days; stockout / lost-sales rate; markdown rate at season-end; inventory turns; freshness (time-on-floor).
- **Decision**: (i) **initial allocation** of a new style across stores; (ii) **replenishment cadence** and cross-store rebalancing; (iii) small-batch "test then scale" production; (iv) markdown timing vs salvage.

- **Randomness**: trend-driven, highly time-varying demand; size–color mix uncertainty; early-week sales signals; returns; occasional lead-time disruptions.

1.2. Decisions under Uncertainty: Why Stochastic Models Matter

Across systems in Example 1.1, Example 1.2 and Example 1.3 , the decision maker must act *before the randomness unfolds*:

- capacity is provisioned, fleets are positioned, and inventory is allocated days or weeks ahead.

IEDA3010

We optimize against **known inputs** (costs, demands, capacities).

IEDA4180 / 4331 / 4420 / 4510 / 4520

The inputs are **random**, and performance is driven by the **distribution** of outcomes, not just the mean.

- Service Engineering (IEDA4180)
- System Risk Management (IEDA4510)
- Quantitative finance (IEDA4331, IEDA4520)
- Pricing and Revenue Management (IEDA4420)

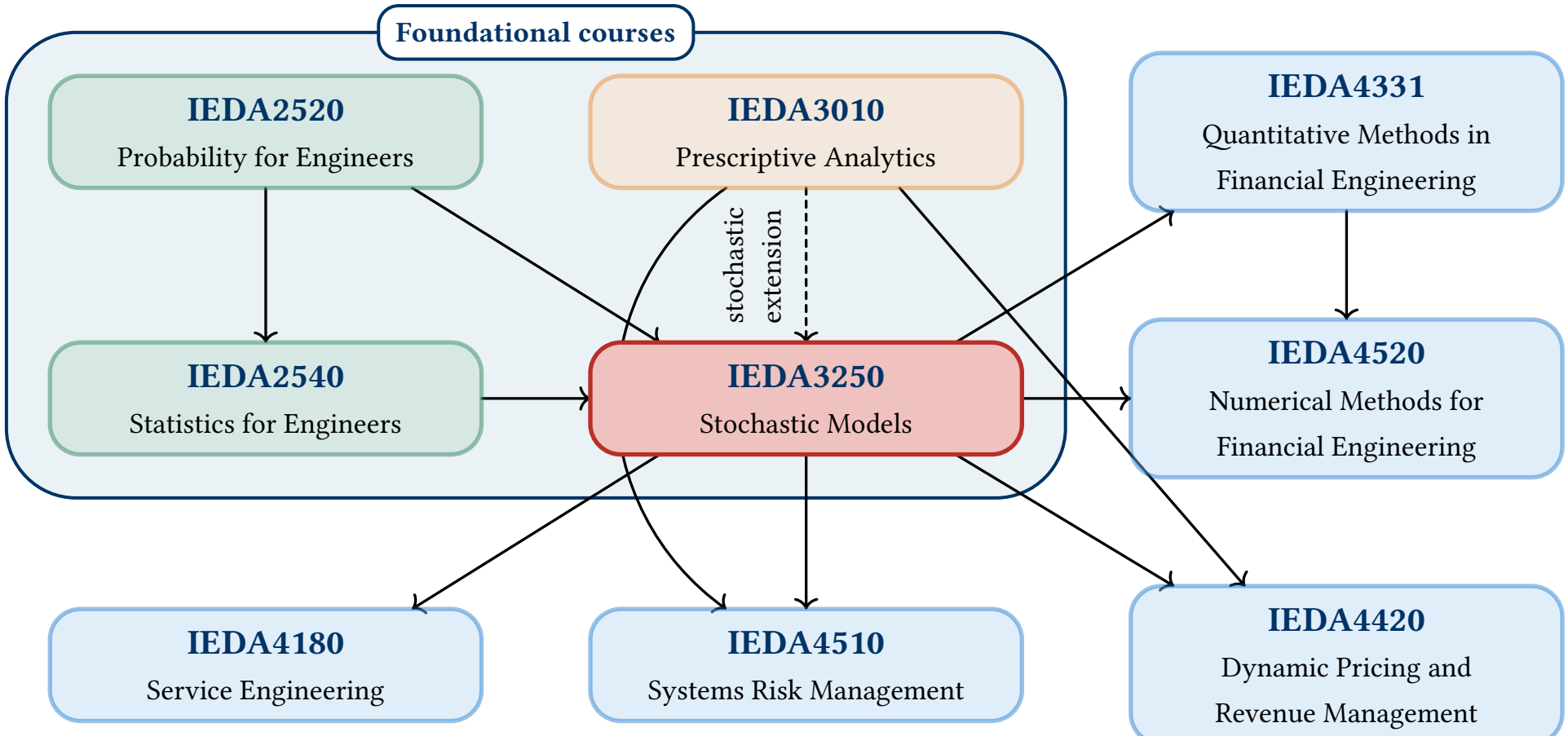


Figure 1.1: IEDA curriculum map: Stochastic Models in context

A deterministic plan says

Example 1.4

- "Average demand is 100/day, so producing 100/day should meet demand."
- "Average processing time is 3 minutes, so capacity of 20 jobs/hour should suffice."
- "Utilization is 80%, so the system should have plenty of slack."

These seem reasonable: the system appears to "keep up" with demand on a "typical day."

⚠ Warning

Planning based on a "average" is fragile, operational performance is driven by **variability**:

- Demand arrives in **bursts**, not as a smooth flow.
- A small fraction of jobs take **much longer** than the mean.
- Rare disruptions (failures, congestion, weather) dominate tail risk.

1.2.1. Case study: Waiting-Time Paradox

Waiting-time / inspection paradox — Example 1.5

You arrive at a bus stop with a schedule that promises a bus every **20 minutes**. You wait… and wait. “Any second now,” you tell yourself, “Surely any second.”—and, at last, the bus arrives.

It feels as if you always arrive one breath after the previous bus has gone.

🙌 Raise your hand if this has happened to you.

Bus wait-time — Poll 1.1

Buses are scheduled every 20 minutes. How long do you expect to wait for the next bus?

A). Less than 10 minutes
B). 10 minutes
C). 10–20 minutes
D). More than 20 minutes

? Question

Consider a simplified scanario where buses arrive **exactly** every 20 minutes. What is your expected waiting time until the next bus arrives?

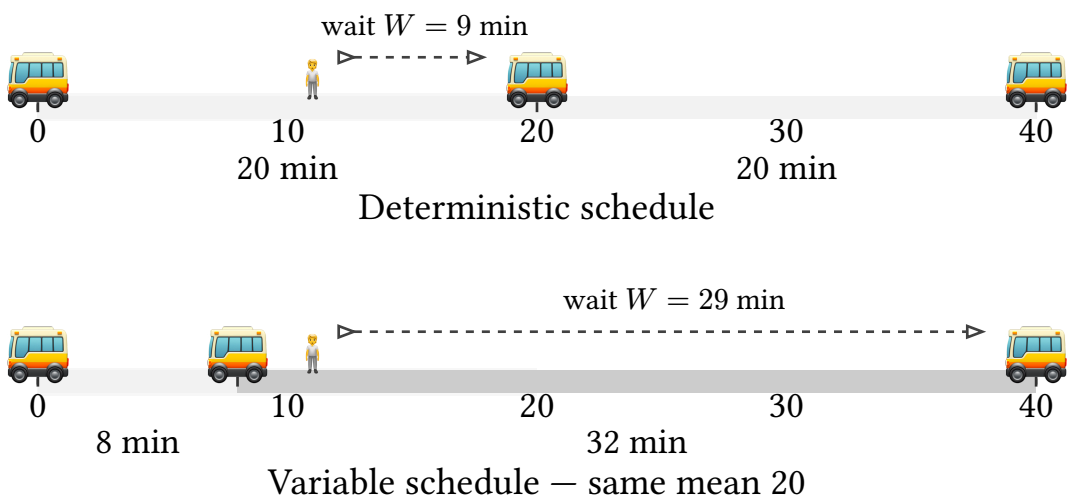


Figure 1.2: Bus waiting: same mean headway, different typical waiting

Is it just bad luck?

How: You are not sampling bus gaps uniformly—you are *sampling time*.

- Long gaps occupy more time on the clock, so you are **more likely to arrive during a long gap** than a short one.
- As a result, the gap you "observe" tends to be longer than a typical gap between buses.

Inspection paradox: when interarrival times are variable, random inspections (your arrival time) are biased toward seeing longer intervals.

When: Passengers experience longer waits *whenever* interarrival times are **irregular**.

i Results (to be proved later in the course)

Suppose buses come "**every 20 minutes**", the mean *experienced* waiting time depends on how *variable* the arrivals are:

- **always larger than 10 = 20/2 minutes**;
- **equals 10 minutes** only when interarrival times are **perfectly deterministic**;
- **equals 20 minutes (!)** for a realistic random bus arrival model.

More on this in *renewal process*.

1.2.2. Case Study: Phantom Traffic Jam

Phantom traffic jam — Example 1.6

You are driving on a highway. The cars ahead suddenly brake. You crawl for **10 minutes**— "An accident," you tell yourself, "Surely an accident."—and then, almost magically, the road clears.

No crash. No cone. No closure. It feels like the jam came from nowhere: a **phantom jam**.

🙌 Raise your hand if this has happened to you.

Why do phantom traffic jams occur? — Poll 1.2

Suppose there is no accidents or bottlenecks, why do phantom traffic jams occur on highways?

A). GPS disrupted.
B). Traffic accident on the opposite side.
C). Trump reelected.
D). A butterfly flapped its wings in Brazil.

Catch the phantom!

- When traffic is dense, tiny random disturbances (one driver hesitates, taps the brake, or leaves a slightly larger gap) get **amplified**.
- The next driver reacts a bit *later* and brakes a bit *harder*; the next one does the same. - Within seconds, a stop-and-go wave forms and *travels backward through the stream.*

Why: The system **breaks** because it is operating near a *critical point*: there is not enough slack to absorb variability. A small fluctuation does not die out; it cascades into a macroscopic jam that persists long after the original disturbance is gone.

My experience: Rubbernecking (a "gapers' delay") Example 1.7

There **is** an accident—but it's on the **opposite** side. My lanes are fully open. Still, traffic on my side suddenly slows because many drivers briefly look over and tap the brakes.



⚠ Warning

A highway designed for **average** traffic flow will inevitably operate near capacity during peak hours. In that regime, small random fluctuations in driving behavior (hesitation, braking, lane changes) amplify into **phantom jams**.

Possible solution: *Reduce variability.* Automated vehicles smooth accelerations and shorten reaction lags, damping disturbances. (Why subway systems rarely "jam": they run on centralized control, scheduled interarrival times, and limited interaction variability.)

! Memorize

When system utilization is high, small variability is not "noise"—it becomes the **driver** of performance, which is why we need stochastic models. (More on this in *queueing theory*.)

1.2.3. Case Study: Pop Mart Blind-Box

Blind-box drop: one decision, uncertain hype **Example 1.8**

A new collectible series is released in "blind boxes": you buy a sealed box without knowing which figure you will get. Some variants are common; a few are **rare**.

On release day, demand can be calm—or it can explode after a viral post. If you under-supply, everything sells out instantly and customers leave frustrated (and scalpers celebrate). If you over-supply, the hype dies, shelves clog, and the "rare chase" feeling evaporates.

🙌 Raise your hand if you have ever tried to get a "drop" item that sold out.

? Question

Suppose last month you sold 10,000 boxes **on average** per drop. For the next drop, should you simply produce 10,000 again?

The “hype tail”

Two drops can have the same **average** demand but feel totally different:

- **Stable world**: demand is consistently near 10,000. Ordering “around 10,000” is usually fine.
- **Hype world**: most drops are modest, but occasionally demand spikes overnight (celebrity post, viral unboxing).
- **What hurts more**?
  - Selling out instantly (lost revenue, angry fans, bad UX).
  - Overstocking (dead inventory, weaker brand momentum).

⚠ Warning

Blind-box products create **two layers of randomness**:

- demand uncertainty (hype, time-of-day spikes, influencer shocks), and
- "mix" uncertainty (which variants customers chase, and how fast the rare ones get depleted).

Planning based on an average can be systematically wrong because what matters is the probability of **extreme** outcomes.

💡 This is a newsvendor problem in collectibles industry

You must commit inventory **before** the demand spike reveals itself. Stochastic thinking forces you to *plan for the distribution* (how often a viral surge occurs), not just the mean.

More on this in *IEDA4420 Dynamic Pricing and Revenue Management*.

1.3. Stochastic Model: Basic Concepts

Any realistic model of a real-world phenomenon must *account for randomness*. In many engineered systems, the quantities we care about—demand, processing times, failures, travel times—are not predictable in advance, but exhibit inherent variation that materially affects performance.

Stochastic model **Definition 1.1**

A **stochastic model** is a mathematical model of a system in which one or more inputs, mechanisms, or state variables are **random**. It specifies:

- the **random quantities** that drive the system and the evolving **state** of the system,
- and the induced **distribution** of performance metrics (e.g., delay, stockouts, reliability).



Random variable — Definition 1.2

A **random variable** X is a rule that converts an outcome of an random experiment into a number (or vector). It lets us treat uncertain quantities like demand as mathematical objects.

Distribution — Definition 1.3

The **distribution** of a random variable describes how likely different values are.

- Discrete case: a probability mass function (PMF) $p(x)$.
- Continuous case: a density (PDF) $f(x)$.
- A universal summary is the CDF: $F(x) = \mathbb{P}(X \leq x)$.

Expectation and variance — Definition 1.4

- **Mean**: $\mathbb{E}[X] = \int_{-\infty}^{\infty} x f(x) \mathrm{d}x$, the long-run average level.
- **Variance**: $\mathrm{Var}(X) = \mathbb{E}\big[(X - \mathbb{E}[X])^2\big]$, how spread out outcomes are around the mean.

Idea

Random variables are the *building blocks* of stochastic models:

- each one encodes an uncertain quantity of interest (e.g., demand, service time, failures).

By **connecting** multiple random variables through the system's operational logic (flows, routing, and orders), we obtain a **time-evolving** description of the system. In other words, uncertainty does not appear as isolated "noise"—it **propagates through time** and across components. *Naturally, we organize a time-indexed family of random variables to describe this evolution. This object is a* ***stochastic process****.*

Stochastic process — Definition 1.5

A **stochastic process** is a family of random variables $\{X(t) : t \in \mathcal{T}\}$ indexed by t, where the index set $\mathcal{T}$ often represents **time**. It models how a random quantity evolves, so we can reason about trajectories, long-run behavior, and event probabilities.

Applications of stochastic processes

Example 1.9

Fields: biology, chemistry, ecology, neuroscience, physics, signal processing, control theory, information theory, computer science, telecommunications, and financial markets.

- Growth of a bacterial population (**over time**)
- Electrical current fluctuating (**over time**) due to thermal noise
- Movement of a gas molecule (**over time**) (*Brownian motion*)
- Arrival of customer orders to an online store (**over time**)
- Queue length / waiting time in a service system (**over time**)
- Inventory level under replenishment and random demand (**over time**)
- Machine up/down state with failures and repairs (**over time**)

In these settings, observations are not viewed as independent data points. Instead, they are **time-coupled** random quantities that evolve with the system's dynamics.

1.4. Overview of Topics

- **Discrete-time stochastic processes** (time in steps: days, jobs, decisions)
  - **Discrete-time Markov chains (DTMCs)**: state transitions each step.
    - Gambler's ruin; random walk with absorbing boundaries.
  - **Martingales**: "fair game" processes and what they imply about predictability.
    - Why no betting system beats a fair game (without edge).
- **Continuous-time stochastic processes** (time on the clock: seconds, minutes, hours)
  - **Counting / renewal processes**: modeling event arrivals over time.
    - Customer orders, calls, ride requests.
  - **Continuous-time Markov chains (CTMCs)**: random transition times between states.
    - Machine failure/repair (birth–death models).
- **Queueing systems** (arrivals + service + congestion dynamics)
  - Hospital/clinic patient flow; call-center waits; platform latency under load.
- **Decision-making under uncertainty**
  - **Markov decision processes (MDPs)**: optimize when the system state is random and evolving.
    - Inventory control and replenishment under random demand.

2. Markov Chains

2.1. Simple Random Walk

Many IE systems evolve by repeating the same pattern:

state today + **random net change** = **state tomorrow**.

The key point is that randomness is not a one-off shock—it arrives again and again, and its effects **accumulate** over time.

Platform growth — Example 2.1

A platform's trajectory can also look like "many small daily wins and losses."

- **State**: X_n = cumulative user base (or a running product "health score").
- **Randomness**: some days bring organic growth; other days bring churn (or incidents).
- **Net change**: ξ_{n+1} = net gain (growth minus churn) on day $n+1$.
- **Question**: Will we eventually "take off" past a target level, or drift down to a critical level?

Gambler's bankroll — Example 2.2

A gambler plays the same game repeatedly. Each round she either wins \$1 or loses \$1.

- **State**: X_n = bankroll after n rounds.
- **Randomness**: each round is a win/loss (think: "good day" vs "bad day").
- **Net change**: $\xi_{n+1} \in \{+1, -1\}$ is the increment in round $n+1$ ($+1$ = win, -1 = loss).
- **Questions**: Starting with i dollars, what is the chance of bankruptcy before reaching N (cashing out)? How long does it take (on average) to either go broke or cash out?

? Question

In these examples, what is the **one number** you would carry forward from today to tomorrow?

- The **state** X_n is a compact summary of "where the system is now,"
- The **net change** ξ_{n+1} determines what happens next.

One-step dynamics

The operational logic is the same across settings:

- We observe the current state X_n.
- A new random increment ξ_{n+1} arrives (demand shock, overrun, churn, etc.).
- The next state is updated by addition: $X_{n+1} = X_n + \xi_{n+1}$.

? Question

There are a few core questions we can ask about "state + random increment" systems:

1. **Is there a systematic drift / trend?**
   - What is the average net change $\mathbb{E}[\xi_n]$?
2. **What happens in the long run?**
   - Does the state X_n tend to grow, shrink, or hover around a typical level?
   - If there is **no** upward/downward drift, what happens eventually?
     - Will we hit zero (bankruptcy, stockout, failure)?
     - Will we reach a high target (takeoff, success)?
3. **How fast do critical events occur?**
   - How long does it take to hit key thresholds (on average, or with high probability)?

2.1.1. Definition of Simple Random Walk

To formalize the repeated "state + random increment" structure, we start with the simplest possible dynamic model: the **simple random walk**.

i Refresher: Independent and identically distributed (i.i.d.)

A sequence of random variables $X_1, X_2, \ldots$ is **i.i.d.** if it satisfies two properties:

- **Identically distributed**: all X_n share the same distribution.
  - *Intuition*: each step is generated by the **same mechanism** (same coin, same demand model, same service-time law).
- **Independent**: knowing past values gives no information about future ones.
  - *Intuition*: the randomness "resets" each time; there is no dependence across periods.

In a simple random walk, the increments $\xi_1, \xi_2, \ldots$ are i.i.d.:

- Each step has the same win/loss probabilities, and one step does not influence the next.

Simple random walk — Definition 2.1

A **simple random walk** is a discrete-time stochastic process $\{X_n : n \geq 0\}$ defined by:

- **Initial state**: $X_0 = 0$.
- **Update rule**:

$$X_{n+1} = X_n + \xi_{n+1}, \tag{2.1}$$

where the increments $\xi_1, \xi_2, \ldots$ are **i.i.d.** with

$$\mathbb{P}(\xi_n = +1) = p, \quad \mathbb{P}(\xi_n = -1) = q \overset{\text{def}}{=} 1 - p. \tag{2.2}$$

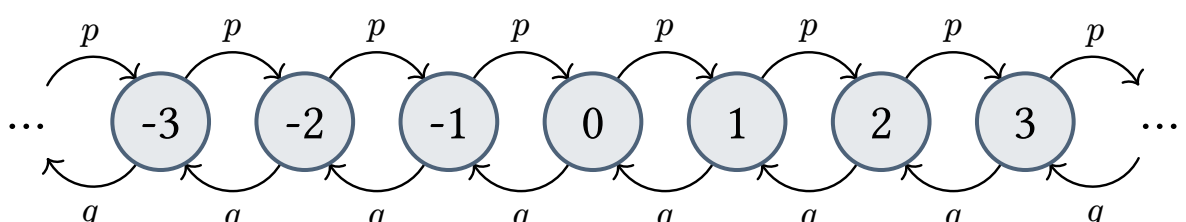


Figure 2.1: State transition diagram of simple random walk.

? In plain language, what does the parameter p control?

Recall that p is the probability of an "up" step. The **trend** of the walk:

- $p > \frac{1}{2}$: "up" steps are more likely (upward drift).
- $p = \frac{1}{2}$: no systematic trend (fair game).
- $p < \frac{1}{2}$: "down" steps are more likely (downward drift).

2.1.2. Properties of Simple Random Walk

To formalize the intuition about trends and long-run behavior, we compute the **mean** and **variance** of the state X_n after n steps.

Simple random walk mean and variance **Proposition 2.1**

For a simple random walk $\{X_n\}$ with step probability p:

- **Mean**: $\mathbb{E}[X_n] = (2p-1)n$.
- **Variance**: $\mathrm{Var}(X_n) = 4pqn = 4p(1-p)n$.

Proof of Proposition 2.1: By linearity of expectation and i.i.d. increments,

$$\mathbb{E}[X_n] = \mathbb{E}\left[\sum_{i=1}^{n} \xi_i\right] = \sum_{i=1}^{n} \mathbb{E}[\xi_i] = n\mathbb{E}[\xi_1].$$

Since $\mathbb{E}[\xi_1] = (+1) \cdot p + (-1) \cdot q = p - q = 2p - 1$, we have $\mathbb{E}[X_n] = n(2p-1)$.

For the variance, since the ξ_i are independent,

$$\mathrm{Var}(X_n) = \mathrm{Var}\left(\sum_{i=1}^{n} \xi_i\right) = \sum_{i=1}^{n} \mathrm{Var}(\xi_i) = n\mathrm{Var}(\xi_1).$$

Now, $\mathbb{E}[\xi_1^2] = (+1)^2 \cdot p + (-1)^2 \cdot q = p + q = 1$, so

$$\mathrm{Var}(\xi_1) = \mathbb{E}[\xi_1^2] - (\mathbb{E}[\xi_1])^2 = 1 - (2p-1)^2 = 4pq = 4p(1-p).$$

Therefore, $\mathrm{Var}(X_n) = n\mathrm{Var}(\xi_1) = 4pqn = 4p(1-p)n$. ∎

Symmetric random walk — Example 2.3

A simple random walk with $p = \frac{1}{2}$ is called a **symmetric random walk**. - 2.1 implies that,

- the symmetric random walk has zero mean $\mathbb{E}[X_n] = 0$;
  - On average, the state tends to stay around zero.
- but variance that **grows linearly** with time: $\mathrm{Var}(X_n) = n$.
  - The position of the state becomes *more and more spread out* over time.

Even when there is no systematic trend, the walk does **not** sit still. Random fluctuations accumulate ($\mathrm{Var}(X_n) = n$), so the state can wander far away and hit critical thresholds.

? Question

What is the distribution of the state?

Because $\xi_1, \xi_2, \ldots, \xi_n \in \{+1, -1\}$, the position $X_n = \sum_{i=1}^{n} \xi_i$ is determined by *how many* "+1 steps" occurred. Let

$$N_n \overset{\text{def}}{=} \sum_{i=1}^{n} \mathbb{1}(\xi_i = +1) \sim \text{Binomial}(n, p),$$

be the number of $+1$ increments among the first n steps. From (2.1), we have

$$X_n = X_{n-1} + \xi_n = (X_{n-2} + \xi_{n-1}) + \xi_n = \sum_{i=1}^{n} \xi_i = (+1)N_n + (-1)(n - N_n) = 2N_n - n.$$

X_n has a shifted-and-scaled binomial distribution

$$\mathbb{P}(X_n = x) = \binom{n}{\frac{n+x}{2}} p^{\frac{n+x}{2}} q^{\frac{n-x}{2}},$$

for values x such that $x \in \{-n, -n+2, \ldots, n\}$; otherwise $\mathbb{P}(X_n = x) = 0$.

Refresher: Central Limit Theorem (CLT)

A sequence of random variables $\xi_1, \xi_2, \ldots$ is **i.i.d.**, each with mean μ and variance $\sigma^2 < \infty$. Then, as $n \to \infty$, the **standardized sum** converges to the standard normal distribution:

$$Z_n \overset{\text{def}}{=} \frac{\sum_{i=1}^{n} \xi_i - n\mu}{\sigma\sqrt{n}} \Rightarrow \mathbb{N}(0,1).$$

Application: The long-run behavior of X_n via CLT — Example 2.4

In a simple random walk with step probability p, we have $\mathbb{E}[\xi_i] = 2p - 1$ and $\mathrm{Var}(\xi_i) = 4pq$. By the Central Limit Theorem, as $n \to \infty$,

$$Z_n \overset{\text{def}}{=} \frac{X_n - n(2p-1)}{2\sqrt{pqn}} \Rightarrow \mathbb{N}(0,1) \quad \text{hence} \quad X_n \approx \mathbb{N}(n(2p-1), 4pqn).$$

? Question

Will a simple random walk hit a fixed target level T (can be positive or negative)?

- In practice, T is often a **threshold** (e.g., 0 for bankruptcy, N for successful cashout).

Hitting Time — Definition 2.2

The **hitting time** τ_a of a level a is the *first time the process touches* a:

$$\tau_a \stackrel{\text{def}}{=} \min\{n \geq 0 : X_n = a\}. \tag{2.3}$$

Intuition: How random walks evolve depends on whether the walk has a **drift**.

Upward drift ($p > \frac{1}{2}$): certainly hits positive targets

Example 2.5

Intuitively, the walk will eventually hit any fixed **positive** target $a > 0$ because it tends to go up. Formally, the mean step is $\mathbb{E}[\xi_1] = p - q = 2p - 1 > 0$.

By the **(strong) law of large numbers**,

$$\frac{X_n}{n} = \left(\frac{1}{n}\right) \sum_{i=1}^{n} \xi_i \Rightarrow \mathbb{E}[\xi_1] = 2p - 1 > 0 \quad \text{as} \quad n \to \infty.$$

This says that, with probability 1, the ratio $\frac{X_n}{n}$ eventually stays close to $2p - 1$. In particular, along almost every sample path, X_n grows roughly linearly and hence $X_n \to \infty$. Therefore, for any fixed $a > 0$, the walk must satisfy $X_n \geq a$ for all sufficiently large n.

Equivalently, the event "$X_n < a$ for all n" has probability 0, i.e.

$$\mathbb{P}(\tau_a < \infty) = 1.$$

Negative drift ($p < \frac{1}{2}$): may never hit a positive target — Example 2.6

When $p < \frac{1}{2}$, the mean step is negative: $\mathbb{E}[\xi_1] = 2p - 1 < 0$. By the law of large numbers,

$$\frac{X_n}{n} \Rightarrow 2p - 1 < 0 \quad \text{as} \quad n \to \infty,$$

so along almost every sample path, X_n eventually trends downward and $X_n \to -\infty$.

Warning

This does **not** mean the walk can never reach a positive target $a > 0$. It can still get lucky early and jump up to a. The point is: there is a **strictly positive** chance it drifts down forever **without ever** reaching a.

Casino game where the house has a small edge — Example 2.7

Think of X_n as your net winnings after n bets in a game with a small house advantage.

- Your goal is to reach a wealth target $a > 0$ before going broke.
- You **might** succeed with an early streak of wins.
- But with negative drift, there is always a nonzero (often substantial) chance you **never** reach a and instead slowly bleed down.

Question: How likely is it to ever hit $a > 0$ when the drift is negative ($p < \frac{1}{2}$)?

Answer (without proof here): The probability is **less than 1** and decays *exponentially* in a:

$$\mathbb{P}(\tau_a < \infty) = \left(\frac{p}{q}\right)^a, \quad q \stackrel{\text{def}}{=} 1 - p.$$

Interpretation: hitting a target **against the drift** requires a sustained run of "up" steps. The farther the target, the more extreme the required streak, so the success probability becomes exponentially small.

Experiment

Discuss the case with **downward drift** ($p < \frac{1}{2}$).

Hitting probabilities under symmetric random walk — Poll 2.1

Consider a **symmetric** random walk ($p = \frac{1}{2}$) starting at $X_0 = 0$, so there is **no systematic drift**.

What is the probability of hitting a fixed target $a > 0$?

A). Probability 0.
B). Probability strictly between 0 and 1.
C). Probability 1.

Hitting time under symmetric random walk (without proofs)

Consider a **symmetric** random walk ($p = \frac{1}{2}$) starting at $X_0 = 0$.

- It hits every fixed level (positive or negative) **with probability 1**:

$$\mathbb{P}(\tau_{+a} < \infty) = \mathbb{P}(\tau_{-a} < \infty) = 1, \quad \forall a > 0.$$

- But it can take a **very long** time; in fact, the expected hitting time is **infinite**:

$$\mathbb{E}[\tau_{+a}] = \mathbb{E}[\tau_{-a}] = \infty, \quad \forall a > 0.$$

Drunkard's walk

A classic interpretation of the **symmetric** random walk ($p = \frac{1}{2}$) is a "drunkard" taking steps left or right at random.

- There is **no drift**: on average the drunkard does not prefer either direction.
- But there is **no stability** either: randomness accumulates, so the position keeps wandering.

Two surprising facts

- **Certain eventual hitting**: the drunkard will eventually stumble onto any fixed location $+a$ or $-a$ with probability 1.
- **No typical hitting time**: while hitting is certain, the expected time to hit a is *infinite*.

IE translation: "No trend" is not the same as "safe." Without drift, the system still explores extremes; buffers and controls determine whether those extremes are acceptable.

The same “drunkard” behaves very differently once you change the dimension:

- **1D**: steps left/right — a drunken ant living on a rope.
- **2D**: steps on a grid — a drunken human wandering in city blocks.

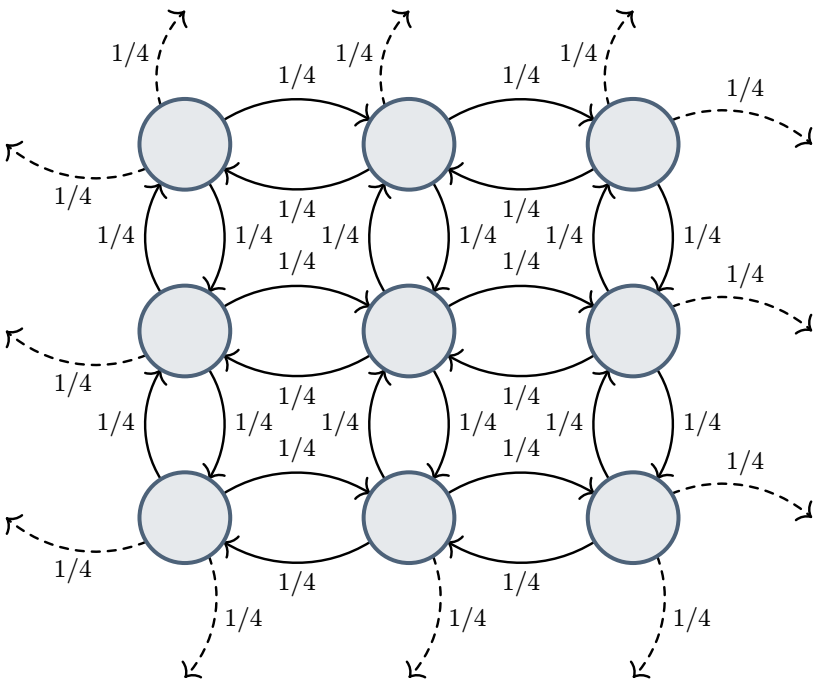


Figure 2.2: State transition diagram of a simple random walk in two-dimensional space.

Fun fact: drunkard's walk in higher dimensions

- **1D (drunken ant) and 2D (drunken human)**:
  - With probability 1, the drunkard returns to the starting point infinitely many times,
  - and will eventually visit any fixed location (given enough time).
  - Such states are called **recurrent**.
- **3D and higher (drunken bird)**:
  - With probability < 1, the drunkard ever returns to the starting point.
  - In fact, there is a nonzero chance they wander off and **never** come back.
  - Such states are called **transient**.

Warning

Adding an extra degree of freedom (moving in 3D instead of 2D) makes "getting lost forever" possible. This is one of the most famous **phase transitions** in stochastic models.

2.2. Discrete-Time Markov Chains

Many engineered systems evolve in **discrete time steps**: at time n, the system updates its *state*, denoted as X_n, according to a (possibly) *random rule*.

- To understand the dynamics of the system, a simplest model would probably be to assume that the X_n are independent random variables, but often such an assumption is clearly unjustified.
  - For instance, starting at some time suppose that X_n represents the price of one share of some security, such as Google, at the end of n additional trading days. Then it certainly seems unreasonable to suppose that the price at the end of day $n + 1$ is independent of the prices on days $n, n - 1, n - 2$ and so on down to day 0.

However, it might be reasonable to suppose that the price at the end of trading day $n + 1$ depends on the previous end-of-day prices only through the price at the end of day n.

In this section, we study a central model for such dynamics: **discrete-time Markov chains (DTMCs)**.

2.2.1. From Random Walk to Markov Chains

In Section 2.1, we saw a canonical example of a discrete-time stochastic process: the **simple random walk**.

✅ Recap: Simple random walk

- **State X_n**:
  - **Discrete time-steps**: evolves in discrete time steps $n = 0, 1, 2, \ldots$,
  - **Discrete state-space**: takes integer values $X_n \in \mathbb{Z}$.
- **Update rule**: $X_{n+1} = X_n + \xi_{n+1}$
  - **Independent increments**: the increments $\xi_1, \xi_2, \ldots$ are i.i.d. and take values in $\{-1, +1\}$.

? Question

What is the **minimum** information you need to predict the distribution of X_{n+1}?

The punchline

Once you know the **current state** X_n, the *detailed past* no longer matters.

- If $X_n = i$, then $\mathbb{P}(X_{n+1} = i+1) = p$ and $\mathbb{P}(X_{n+1} = i-1) = q \overset{\text{def}}{=} 1-p$, regardless of how you got to i.

This "memoryless" property is the defining feature of the so-called **Markov chains**:

> **Markov property**: *Given the present, the future does not depend on the detailed past.*

Refresher: conditional probability (what does $\mathbb{P}(A \mid B)$ mean?)

Conditional probability formalizes "probability **given** that we know B happened."

Events. For events A and B with $\mathbb{P}(B) > 0$, define

$$\mathbb{P}(A \mid B) \overset{\text{def}}{=} \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}.$$

Two immediate consequences (use these constantly):

- **Multiplication rule**:

$$\mathbb{P}(A \cap B) = \mathbb{P}(A \mid B)\mathbb{P}(B).$$

- **Law of total probability** (partition by disjoint cases $B_1, \ldots, B_m$):

$$\mathbb{P}(A) = \sum_{k=1}^{m} \mathbb{P}(A \mid B_k)\mathbb{P}(B_k).$$

Conditional probability for random variables: "conditional distribution"

For discrete random variables X, Y, conditioning on "$Y = y$" means conditioning on the event $\{Y = y\}$. If $\mathbb{P}(Y = y) > 0$, define

$$\mathbb{P}(X = x \mid Y = y) \overset{\text{def}}{=} \frac{\mathbb{P}(\{X = x\} \cap \{Y = y\})}{\mathbb{P}(\{Y = y\})}.$$

So, **knowing $Y = y$ changes the distribution of X**.

For the simple random walk, the Markov property can be expressed as:

$$\begin{aligned}
&\mathbb{P}(X_{n+1} = j \mid X_n = i, X_{n-1} = i_{n-1}, \ldots, X_0 = i_0) \\
&= \mathbb{P}(X_{n+1} = j \mid X_n = i) \\
&= \begin{cases} p \text{ if } j = i + 1 \\ q \text{ if } j = i - 1 \\ 0 \text{ otherwise.} \end{cases}
\end{aligned}$$

Let $\mathcal{S}$ denote the **state space** of the process (the set of all possible states).

Discrete-Time Markov Chain (DTMC) — Definition 2.3

A discrete-time stochastic process (c.f. Definition 1.5) $\{X_0, X_1, X_2, ...\}$ is called a **discrete-time Markov chain (DTMC)** if, for every $n \geq 0$ and all states $i_0, i_1, ..., i, j \in \mathcal{S}$,

$$\mathbb{P}(X_{n+1} = j \mid X_n = i, X_{n-1} = i_{n-1}, ..., X_0 = i_0) = \mathbb{P}(X_{n+1} = j \mid X_n = i).$$

We focus on simpler *time-homogeneous* cases where the step mechanism does not change with n.

Time-homogeneous DTMC — Definition 2.4

A DTMC is called **time-homogeneous** if there is a table of **one-step transition probabilities**

$$P_{ij} \overset{\text{def}}{=} \mathbb{P}(X_{n+1} = j \mid X_n = i), \quad i, j \in \mathcal{S}, \quad \text{that does not depend on } n.$$

The matrix $P = \left(P_{ij}\right)_{\{ij \in \mathcal{S}\}}$ is called the **transition matrix** of the DTMC.

⚠ Markov is about the state representation

Not all stochastic processes are Markov **in the variables you happen to observe**.

Example (CGA is not "memoryless"): Your CGA at the end of year n certainly matters for predicting next year's CGA, but it is **not** enough information by itself:

- Two students can have the same current CGA but very different **accumulated credits**, so the same future semester grade will move their CGA by different amounts.
- Therefore, the distribution of next year's CGA depends on more than just the current CGA—it depends on **how you got there** (via credit history).

Key lesson: A process can fail to be Markov because the "state" is missing information.

How to fix it (state augmentation): If you define the state as

(1) **total grade points so far** and (2) **total credits so far**,

then the next update depends only on this *augmented state* plus the next semester's outcome.

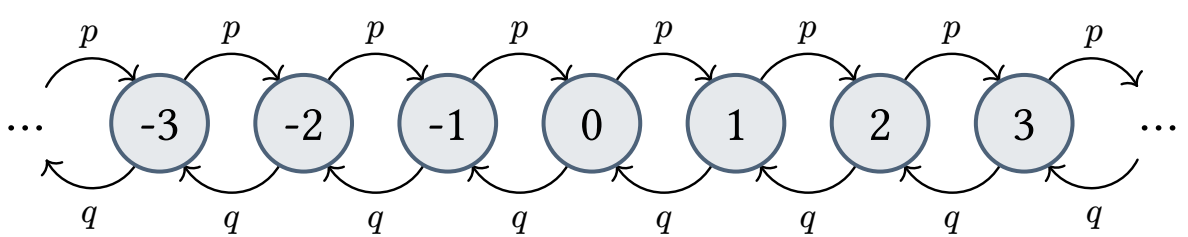
Random walk as a DTMC

Example 2.8

- **State space**: $\mathcal{S} = \mathbb{Z}$.
- **Randomness**: walk left with probability p or right with probability q.
- **Transitions**: from state i,
  - go to $i+1$ with probability p,
  - go to $i-1$ with probability $q \overset{\text{def}}{=} 1-p$.

- **Transition matrix**:

$$P = \begin{array}{c} \\ -2 \\ -1 \\ 0 \\ 1 \\ 2 \end{array} \begin{array}{c} \begin{array}{ccccccc} -3 & -2 & -1 & 0 & 1 & 2 & 3 \end{array} \\ \begin{pmatrix} \ddots & \ddots & \ddots & \ddots & \ddots & & \\ \ddots & q & 0 & p & 0 & 0 & \\ \ddots & 0 & q & 0 & p & 0 & \ddots \\ & 0 & 0 & q & 0 & p & \ddots \\ & & \ddots & \ddots & \ddots & \ddots & \ddots \end{pmatrix} \end{array},$$



! Memorize

The row of the transition matrix P each sum to 1:

$$\sum_{j \in \mathcal{S}} P_{ij} = 1, \quad \forall i \in \mathcal{S}.$$

This is because from any state i, the process must transition to **some** state j in the next step. Hence, the probabilities over all possible j must add up to 1.

Stochastic matrix — Definition 2.5

A (right / row) **stochastic matrix** is a *square matrix* of nonnegative real numbers, with each row summing to 1. The transition matrix P of a DTMC is always a stochastic matrix.

Machine up/down as a DTMC

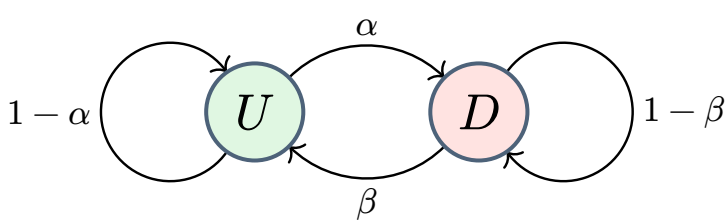
Example 2.9

- **State**: $X_n \in \{U, D\}$ = machine is **Up** / **Down** at day n.
- **Randomness**: If the machine is down, a technician performs repairs with probability α; otherwise, the machine may fail with probability β.
- **Transitions** (time-homogeneous):
  - $U \to D$ w.p. α (failure),
  - $D \to U$ w.p. β (repair),
  - otherwise it stays.

- **Transition matrix**:

$$P = \begin{array}{c} \\ U \\ D \end{array} \begin{array}{c} \begin{array}{cc} U & D \end{array} \\ \begin{pmatrix} 1-\alpha & \alpha \\ \beta & 1-\beta \end{pmatrix} \end{array}$$

- **Transition diagram:**



Remarks:

- Again, we see that the rows of the transition matrix sum to 1.
- A state can have transitions to itself (e.g., machine stays up or down).

? Question

In the machine example, what if the failure risk depends on "days since last maintenance"?

⚠ $\{U, D\}$ is no longer Markov

If we keep the state space $\mathcal{S} = \{U, D\}$, then the one-step transition $\mathbb{P}\big(X_{n+1} = D \,\big|\, X_n = U\big)$ is **not** well-defined, as we would have to know **how long** the machine has been running since the last maintenance.

In other words, the transition probabilities depend on the **history** (an "age" variable), not just on whether the machine is currently up or down.

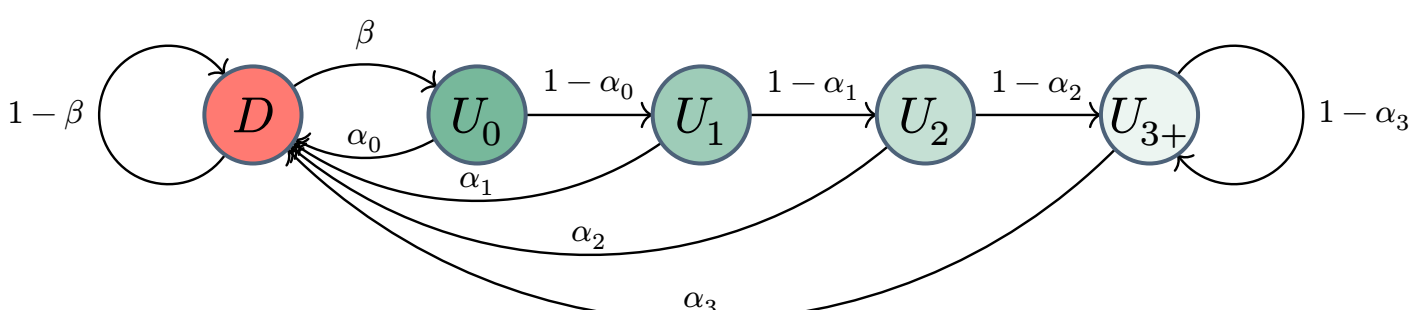
Augment the state to restore the Markov property

Add a clock-like variable: "days since last maintenance."

A simple augmented state space is $\mathcal{S}^+ = \{U_0, U_1, U_2, U_3, D\}$, where

- U_k: machine is **Up** and it has been k days since the last maintenance,
- D: machine is **Down**.

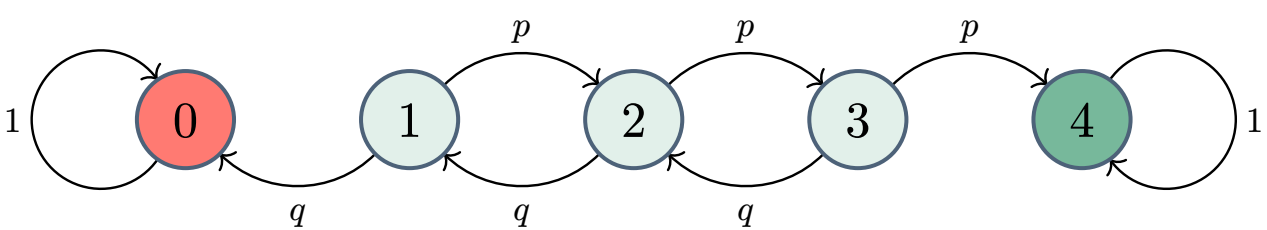
With this augmentation, the next-day behavior depends only on the **current** augmented state, so the process becomes Markov again. The transition diagram is illustrated below:



Gambler's ruin

Example 2.10

Consider a gambler who, at each play of the game, either wins \$1 with probability p or loses \$1 with probability $q = 1 - p$. If we suppose that our gambler quits playing either when he goes broke or he attains a fortune of \$4, then the gambler's fortune is a Markov chain with state space $\mathcal{S} = \{0, 1, 2, 3, 4\}$, where state i represents $\$i$.



$$P = \begin{array}{c} \\ 0 \\ 1 \\ 2 \\ 3 \\ 4 \end{array} \begin{array}{c} \begin{array}{ccccc} 0 & 1 & 2 & 3 & 4 \end{array} \\ \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ q & 0 & p & 0 & 0 \\ 0 & q & 0 & p & 0 \\ 0 & 0 & q & 0 & p \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix} \end{array}$$

Absorbing states

Definition 2.6

A state that, once entered, is never left is called an absorbing state. In the gambler's ruin example, states 0 and 4 are absorbing states.

Inventory chain: replenishment with a reorder point — Example 2.11

Let X_n be the **end-of-day** inventory after day n, and let D_{n+1} be the demand on day $n+1$.

(s, S)-policy:

- If $X_n \le s$, we place an order so that the **start-of-day** inventory becomes S.
- If $X_n > s$, we order nothing.
- Assume replenishment order arrives before demand on day $n+1$ is realized.

The dynamics can be written compactly as

$$X_{n+1} = \begin{cases} (X_n - D_{n+1})^+ & \text{if } X_n > s \\ (S - D_{n+1})^+ & \text{if } X_n \le s. \end{cases}, \quad \text{where} \quad x^+ \overset{\text{def}}{=} \max\{x, 0\} = \begin{cases} x & \text{if } x > 0 \\ 0 & \text{if } x \le 0 \end{cases}.$$

So $(X_n)_{n \ge 0}$ is a DTMC on the state space $\mathcal{S} = \{0, 1, \ldots, S\}$.

Suppose $s = 1$ and $S = 5$, and demand takes values $0, 1, 2, 3$ with

$$\mathbb{P}(D_{n+1} = 0) = 0.3, \quad \mathbb{P}(D_{n+1} = 1) = 0.4, \quad \mathbb{P}(D_{n+1} = 2) = 0.2, \quad \mathbb{P}(D_{n+1} = 3) = 0.1.$$

- **States**: $\mathcal{S} = \{0, 1, 2, 3, 4, 5\}$.
- **One-step transition matrix**:

$$P = \begin{array}{c} \\ 0 \\ 1 \\ 2 \\ 3 \\ 4 \\ 5 \end{array} \begin{array}{c} \begin{array}{cccccc} 0 & 1 & 2 & 3 & 4 & 5 \end{array} \\ \begin{pmatrix} 0 & 0 & 0.1 & 0.2 & 0.4 & 0.3 \\ 0 & 0 & 0.1 & 0.2 & 0.4 & 0.3 \\ 0.3 & 0.4 & 0.3 & 0 & 0 & 0 \\ 0.1 & 0.2 & 0.4 & 0.3 & 0 & 0 \\ 0 & 0.1 & 0.2 & 0.4 & 0.3 & 0 \\ 0 & 0 & 0.1 & 0.2 & 0.4 & 0.3 \end{pmatrix} \end{array}$$

Why rows 0 and 1 look the same

If $X_n \in \{0, 1\}$, the policy replenishes to 5 **before** demand arrives. Therefore, starting from 0 or 1 has the same next-day distribution as starting from 5.

Question

If we start with $X_0 = 5$, what is the distribution after 1 day and 2 days?

Define the state distribution row vector

$$\mu^{(n)} \overset{\text{def}}{=} \left(\mu_j^{(n)}\right)_{j=0,1,2,3,4,5}, \quad \mu_j^{(n)} \overset{\text{def}}{=} \mathbb{P}(X_n = j).$$

With $X_0 = 5$ we have

$$\mu^{(0)} = (0, 0, 0, 0, 0, 1).$$

By the Kolmogorov update $\mu^{(n+1)} = \mu^{(n)} P$,

$$\mu^{(1)} = \mu^{(0)} P = (0, 0, 0.1, 0.2, 0.4, 0.3).$$

and one more step gives

$$\mu^{(2)} = \mu^{(1)} P = (0.05, 0.12, 0.22, 0.28, 0.24, 0.09).$$

! Memorize

This is the DTMC workflow in one line: **model the policy** $\Rightarrow$ **write** P $\Rightarrow$ **forecast via** $\mu^{(n)} = \mu^{(0)} P^n$.

? Question

Profit application: Suppose the store earns \$12 per unit sold and pays \$2 per unit carried (per day).

- What is the **long-run** average profit per day under $(s, S) = (1, 5)$?
- How should we choose (s, S) to maximize long-run profit?

2.2.2. Chapman–Kolmogorov Equations: Multi-Step Transitions

? Question

We already know the **one-step** rule:

$$P_{ij} = \mathbb{P}(X_{n+1} = j \mid X_n = i) \tag{2.4}$$

What if you need a **two-step** forecast: $\mathbb{P}(X_{n+2} = j \mid X_n = i)$?

The intermediate state trick

To go from i to j in two steps, the chain must pass through **some** intermediate state k at time $n+1$.

Trick: Use the law of total probability, and then use the Markov property to "cut" the dependence.

Two-step transition probabilities: For any states $i, j \in \mathcal{S}$, and any intermediate step $k \in \mathcal{S}$,

$$\begin{aligned}
\mathbb{P}(X_{n+2} = j \mid X_n = i) &= \sum_{k \in \mathcal{S}} \mathbb{P}(X_{n+2} = j, X_{n+1} = k \mid X_n = i) \\
&= \sum_{k \in \mathcal{S}} \mathbb{P}(X_{n+2} = j \mid X_{n+1} = k, X_n = i) \mathbb{P}(X_{n+1} = k \mid X_n = i) \\
&= \sum_{k \in \mathcal{S}} \mathbb{P}(X_{n+2} = j \mid X_{n+1} = k) \mathbb{P}(X_{n+1} = k \mid X_n = i) \\
&= \sum_{k \in \mathcal{S}} P_{ik} P_{kj}.
\end{aligned}$$

- The first equality is the **law of total probability**, summing over all possible intermediate states k at time $n + 1$.
- The second equality uses the definition of conditional probability.
- The third equality uses the **Markov property** to eliminate dependence on $X_n = i$ when conditioning on $X_{n+1} = k$.

Two-step transitions by matrix squaring

Proposition 2.2

If P is the one-step transition matrix, then the two-step transition matrix is

$$P^{(2)} = P \cdot P = P^2 = \begin{bmatrix} P_{11} & P_{12} & \dots & P_{1J} \\ P_{21} & P_{22} & \dots & P_{2J} \\ \vdots & \vdots & \ddots & \vdots \\ P_{I1} & P_{I2} & \dots & P_{IJ} \end{bmatrix} \cdot \begin{bmatrix} P_{11} & P_{12} & \dots & P_{1J} \\ P_{21} & P_{22} & \dots & P_{2J} \\ \vdots & \vdots & \ddots & \vdots \\ P_{I1} & P_{I2} & \dots & P_{IJ} \end{bmatrix},$$

with entries $P_{ij}^{(2)} = \sum_{k \in \mathcal{S}} P_{ik} P_{kj}$.

Sanity check: random walk returns in two steps

Example 2.12

For a simple random walk on $\mathbb{Z}$ with $\mathbb{P}(\xi = +1) = p$ and $\mathbb{P}(\xi = -1) = q$, starting from 0 the only way to return to 0 in 2 steps is "up then down" or "down then up".

$$\mathbb{P}(X_2 = 0 \mid X_0 = 0) = pq + qp = 2pq.$$

? Question

How to compute the n-step transition probabilities $\mathbb{P}(X_{m+n} = j \mid X_m = i)$?

n-step transition probabilities — Poll 2.2

Given the one-step transition matrix P, how to compute the n-step transition matrix $P^{(n)}$?

A). P^n (matrix multiplication).
B). nP (scalar multiplication).
C). $P + P + \ldots + P$ (matrix addition).

n-step transition probabilities

Definition 2.7

For a time-homogeneous DTMC on state space $\mathcal{S}$, define

$$P_{ij}^{(n)} \overset{\text{def}}{=} \mathbb{P}(X_{m+n} = j \mid X_m = i), \quad n = 0, 1, 2, \ldots$$

which does not depend on m.

The **Chapman–Kolmogorov equations** provide a method for computing n-step transition probabilities.

Chapman–Kolmogorov equations

Definition 2.8

For any $m, n \geq 0$ and any states i, j,

$$P_{ij}^{(m+n)} = \sum_{k \in \mathcal{S}} P_{ik}^{(m)} P_{kj}^{(n)}. \tag{2.5}$$

Proof of Equation (2.5): Fix an arbitrary start time r and condition on the **intermediate** state X_{r+m}. Using the law of total probability,

$$\begin{aligned} P_{ij}^{(m+n)} &= \mathbb{P}(X_{r+m+n} = j \mid X_r = i) \\ &= \sum_{k \in \mathcal{S}} \mathbb{P}(X_{r+m+n} = j, X_{r+m} = k \mid X_r = i) \\ &= \sum_{k \in \mathcal{S}} \mathbb{P}(X_{r+m+n} = j \mid X_{r+m} = k, X_r = i) \mathbb{P}(X_{r+m} = k \mid X_r = i) \\ &= \sum_{k \in \mathcal{S}} \mathbb{P}(X_{r+m+n} = j \mid X_{r+m} = k) \mathbb{P}(X_{r+m} = k \mid X_r = i), \end{aligned}$$

where the last line follows from the *Markov property*: given $X_{r+m} = k$ the future does not depend on the earlier history. Finally, by time-homogeneity (shift-invariance),

$$\mathbb{P}(X_{r+m} = k \mid X_r = i) = P_{ik}^{(m)}, \quad \mathbb{P}(X_{r+m+n} = j \mid X_{r+m} = k) = P_{kj}^{(n)}.$$

Substituting these into the sum gives (2.5). ∎

! Memorize

Let P be the one-step transition matrix. By Chapman–Kolmogorov equations, we have the matrix multiplication rule:

$$P^{(m+n)} = P^{(m)} \cdot P^{(n)} = \begin{bmatrix} P_{11}^{(m)} & P_{12}^{(m)} & \cdots & P_{1J}^{(m)} \\ P_{21}^{(m)} & P_{22}^{(m)} & \cdots & P_{2J}^{(m)} \\ \vdots & \vdots & \ddots & \vdots \\ P_{I1}^{(m)} & P_{I2}^{(m)} & \cdots & P_{IJ}^{(m)} \end{bmatrix} \cdot \begin{bmatrix} P_{11}^{(n)} & P_{12}^{(n)} & \cdots & P_{1J}^{(n)} \\ P_{21}^{(n)} & P_{22}^{(n)} & \cdots & P_{2J}^{(n)} \\ \vdots & \vdots & \ddots & \vdots \\ P_{I1}^{(n)} & P_{I2}^{(n)} & \cdots & P_{IJ}^{(n)} \end{bmatrix}, \quad \forall m, n \geq 0.$$

In particular, taking $n = 1$ and iterating gives the familiar "matrix power" representation:

$$P^{(n)} = P^{(n-1)} \cdot P = P^{(n-2)} \cdot P \cdot P = \cdots = P^n, \quad n = 0, 1, 2, \ldots$$

Once you know P, every multi-step transition probability is an entry of P^n. In this sense, the full evolution of a time-homogeneous DTMC is encoded in its one-step transition matrix P.

? Question

So far, we have focused on **point-to-point** transitions. Often in application, we don't care about the exact path along which the system evolves, but rather the **distribution** of the state.

- **Example:** in gambling, we do not care how we got to our current fortune, only (the probability of) how much money we have now.
- **Question:** How does the distribution of the state evolve day by day?

Define the **state distribution row vector**

$$\mu^{(n)} \overset{\text{def}}{=} \left(\mu_j^{(n)}\right)_{j\in\mathcal{S}}, \quad \mu_j^{(n)} \overset{\text{def}}{=} \mathbb{P}(X_n = j), \quad n = 0, 1, 2, \ldots$$

Discrete-time Kolmogorov forward equation — Theorem 2.1

Let $(X_n)_{n\geq 0}$ be a time-homogeneous DTMC with one-step transition matrix P. Then the one-step distribution update is

$$\mu^{(n+1)} = \mu^{(n)} P, \quad n \geq 0,$$

equivalently, componentwise,

$$\mu_j^{(n+1)} = \sum_{i \in \mathcal{S}} \mu_i^{(n)} P_{ij}, \quad j \in \mathcal{S}.$$

Iterating yields the n-step distribution

$$\mu^{(n)} = \mu^{(0)} P^n, \quad n = 0, 1, 2, \ldots$$

(with P^0 the identity matrix).

Proof of Theorem 2.1: Fix any $j \in \mathcal{S}$. By the law of total probability,

$$\mu_j^{(n+1)} = \mathbb{P}(X_{n+1} = j) = \sum_{i \in \mathcal{S}} \mathbb{P}(X_{n+1} = j, X_n = i).$$

Then factor the joint probability:

$$\mathbb{P}(X_{n+1} = j, X_n = i) = \mathbb{P}(X_{n+1} = j \mid X_n = i)\mathbb{P}(X_n = i) = P_{ij}\mu_i^{(n)}.$$

Summing over i gives

$$\mu_j^{(n+1)} = \sum_{i \in \mathcal{S}} \mu_i^{(n)} P_{ij},$$

which is exactly the matrix identity $\mu^{(n+1)} = \mu^{(n)} P$. Iterating over n yields

$$\mu^{(n)} = \mu^{(n-1)} P = \mu^{(n-2)} P^2 = \cdots = \mu^{(0)} P^n.$$

Machine up/down (cont.) Example 2.13

State space $\mathcal{S} = \{U, D\}$ with ordering (U, D).

Transitions:

- $U \to D$ w.p. α, so $U \to U$ w.p. $1 - \alpha$.
- $D \to U$ w.p. β, so $D \to D$ w.p. $1 - \beta$.

$$P = \begin{pmatrix} 1-\alpha & \alpha \\ \beta & 1-\beta \end{pmatrix}$$

? Question

Suppose the machine is **up** in day 0, i.e., $\mu^{(0)} = (1, 0)$. What is the probability it is **down** after 1 day? after 2 days?

After 1 day:

$$\mu^{(1)} = \mu^{(0)}P = (1,0)P = (1-\alpha, \alpha) \quad \Rightarrow \quad \mathbb{P}(X_1 = D) = \mu^{(1)}\mathbf{1}_D = \alpha$$

so .

After 2 days:

$$\mu^{(2)} = \mu^{(1)}P = (1,0)P^2, \text{where } P^2 = \begin{pmatrix} (1-\alpha)^2 + \alpha\beta & \alpha(2-\alpha-\beta) \\ \beta(2-\alpha-\beta) & (1-\beta)^2 + \alpha\beta \end{pmatrix}.$$

Thus,

$$\mathbb{P}(X_2 = D) = \mu^{(0)}P^2\mathbf{1}_D = (1,0)P^2\mathbf{1}_D = (1-\alpha)\alpha + \alpha(1-\beta) = \alpha(2-\alpha-\beta).$$

Here, $\mathbf{1}_D$ is the *column vector* with 1 in the D component and 0 elsewhere.

- What if the distribution at time 0 is $\mu^{(0)} = \left(\frac{1}{2}, \frac{1}{2}\right)$?

2.2.3. Limiting behavior of DTMCs

asdasd